Note on: *Pricing Index-Linked Hybrids under a Local Volatility Market Model*

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Reading notes

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One-paragraph summary

A short note proposing a "market model" for pricing a cash-settled swaption whose strike depends on the level of an equity index at the exercise date T: payoff $\widehat{A}_T \cdot (\theta(S_T - U_T))^+$ where $\widehat{A}_T$ is the cash annuity, S_T the forward swap rate, and U_T the index forward. The key trick is to switch from the swaption measure to the T-forward measure: the cash annuity then becomes a deterministic function of the convexity-adjusted forward swap rate $F_{TT} = S_T$, so only the joint distribution of (F, U) matters. Both F and U are modelled as local-volatility martingales under the T-forward measure (calibratable individually to vanilla swaption and vanilla index option markets), with an exogenously specified correlation ρ. The framework keeps every dynamics "market observable": F_{0T} from CMS quotes / static replication, σ_F from swaptions, σ_U from index options, ρ from historical data.

1 Setup and notation

Paper	House	Meaning
U_t	—	Spot index level
U_{tT}	—	Forward price of the index at t for delivery T
S_t	R_t^{swp}	Forward swap rate at t for tenor $T_1, \ldots, T_n$
A_t	A_t	Risk-free annuity, $\sum_i \alpha_i D_{t,T_i}$
$\widehat{A}_T$	$\widehat{A_T}$	Cash annuity (flat-curve discount on S_T)
θ	$\theta \in \{+1, -1\}$	Call $(+1)$ / put (-1) flag
$T = T_0$	T	Swaption exercise date
$T_1, \ldots, T_n$	T_i	Underlying swap payment dates
D_{tT_i}	D_{t,T_i}	Risk-free DF
F_{tT}	F_{tT}	$\mathbb{E}_t^T[S_T]$, convexity-adjusted forward swap rate
σ_F, σ_U	—	Local volatility functions
W^F, W^U	—	Driving BMs under the T-forward measure
ρ	ρ	Instantaneous correlation
K	K	Vanilla swaption / index option strike

2 Key equations

Cash annuity (flat-curve proxy):

$$\widehat{A}_T \;=\; \sum_{i=1}^{n} \frac{\alpha_i}{(1+\alpha_i S_T)^{\mathrm{yf}(T,T_i)}}, \tag{1}$$

with $\alpha_i = \mathrm{yf}(T_{i-1}, T_i)$. *Says:* the cash annuity replaces market discount factors with synthetic ones computed from a flat curve at S_T. Standard for cash-settled swaptions in the European market. By construction, $\widehat{A}_T$ is a function of S_T alone.

Index-linked cash-settled swaption payoff:

$$\widehat{A}_T\,(\theta(S_T - U_T))^+. \tag{2}$$

Says: a cash-settled swaption with strike replaced by the realised level of the index U_T.

Generic numeraire-pair PV:

$$V_t \;=\; N_t\,\mathbb{E}_t^N\Big[\theta\,\widehat{A}_T\,(\theta(S_T - U_T))^+/N_T\Big]. \tag{3}$$

Switch to T-forward measure: With $F_{tT} \equiv \mathbb{E}_t^T[S_T]$ the convexity-adjusted forward swap rate and noting $F_{TT} = S_T$,

$$V_t \;=\; D_{t,T}\,\mathbb{E}_t^T\Big[\widehat{A}_T\,(\theta(F_{TT} - U_{TT}))^+\Big]. \tag{4}$$

Says: the change of numeraire reduces the joint problem to specifying the bivariate distribution of (F_{TT}, U_{TT}) under the T-forward measure; no need to model $\widehat{A}_T/A_T$ separately because $\widehat{A}_T$ is a function of $S_T = F_{TT}$.

Vanilla calibration targets (priced under the same model):
vanilla swaption (cash-settled):

$$V_t^{\mathrm{swp}} \;=\; D_{t,T}\,\mathbb{E}_t^T\Big[\widehat{A}_T\,(\theta(F_{TT} - K))^+\Big], \tag{5}$$

CMS cap / floor on the convexity-adjusted rate (often not liquid):

$$V_t^{\text{cms}} = D_{t,T}\,\mathbb{E}_t^T\Big[\big(\theta(F_{TT}-K)\big)^+\Big], \tag{6}$$

vanilla index option:

$$V_t^{\text{idx}} = D_{t,T}\,\mathbb{E}_t^T\Big[\big(\theta(U_{TT}-K)\big)^+\Big]. \tag{7}$$

Local-volatility dynamics under T-forward measure:

$$\frac{\mathrm{d}F_{tT}}{F_{tT}} = \sigma_F(t, F_{tT})\ \mathrm{d}(W^F)_t^T, \tag{8}$$

$$\frac{\mathrm{d}U_{tT}}{U_{tT}} = \sigma_U(t, U_{tT})\ \mathrm{d}(W^U)_t^T, \tag{9}$$

$$\mathrm{d}\langle (W^F)^T, (W^U)^T\rangle_t = \rho\ \mathrm{d}t. \tag{10}$$

Says: both forwards are martingales under $\mathbb{Q}^T$, each with its own Dupire-style local-volatility function. The correlation ρ is exogenous to the calibration of marginals.

3 Derivations & commentary

Commentary

The structural simplification is that $\widehat{A}_T$ is, by definition (1), a deterministic function of S_T. So switching to T-forward leaves a problem entirely in (F_{TT}, U_{TT}). Compared with traditional hybrid approaches (short-rate + equity, e.g. G2++/BS or Heston-HW), this is much lighter: no unobservable short-rate factors, no joint stochastic-volatility specification, no FX-style hybrid correlation matrix.

Commentary

The trade-off is honesty about correlation. ρ between F_{tT} and U_{tT} is not directly observable: there is no liquid product whose price depends only on it. The paper's suggestions are (i) interpolate forward swap rates along zero curves of an index series as a proxy for the convexity-adjusted rate, then estimate historical ρ with U_{tT}; or (ii) use Black's ATM approximation on historical forward swap rates to back out a historical F_{tT} time series. Both are heuristics. As with the CMASW paper, the model honestly admits that correlation is the soft spot.

Commentary

The convexity-adjusted CMS rate F_{0T} is treated as a market input. Practically, this is read off CMS spread quotes or a CMS interpolator (often SABR-based) on the rates desk. The note's calibration recipe is therefore:

1. F_{0T} from CMS market.
2. $\sigma_F(\cdot,\cdot)$ from swaption smile via (5) (Dupire on the cash-swaption-implied marginal of F).
3. $\sigma_U(\cdot,\cdot)$ from index option smile via (7) (standard Dupire on the index).
4. ρ exogenous, ideally historical.

Commentary

The paper says the framework extends to "other multi-asset hybrid payoffs of European type". The key ingredients are: (i) the payoff must be European; (ii) the cash component must reduce to a function of a single underlying after the measure change; (iii) the remaining underlyings need vanilla-option markets for marginal calibration. Cliquets, autocallables and other path-dependent hybrids are out of scope.

4 Validation

Validation

This is a *specification* of mathematical objects the paper defines and the numerical values they should take. It says nothing about how those objects are implemented or invoked.
Quantities the paper defines:

- Cash annuity $\widehat{A}_T$ via (1), a deterministic function of S_T.
- Index-linked cash-swaption payoff at T via (2).
- T-forward measure pricing formula (4); convexity-adjusted forward $F_{tT} = \mathbb{E}_t^T[S_T]$.
- Three vanilla-option pricing formulas in the same framework (5)–(7), used as calibration targets.
- Joint local-volatility dynamics (8)–(9) with instantaneous correlation ρ.

Canonical numerical case: the paper provides no worked numerical example. Suggested test bench: a 1Y expiry on a 10Y cash-settled swaption (10Y tenor, annual payments) struck against a major equity index (e.g. S&P 500 or EuroStoxx 50). Use F_{0T} read off a CMS interpolator, swaption ATM vol $\sim 30\%$, index ATM vol $\sim 20\%$, $\rho \in \{-0.3, 0.0, +0.3\}$ as a sensitivity grid.
Expected numerical values: none printed in the paper.
Equation $\leftrightarrow$ quantity map:

Paper eq.	Symbol	Quantity
(1)	$\widehat{A}_T$	cash annuity as deterministic fn of S_T
(2)	$\widehat{A}_T(S_T - U_T)^+$	index-linked cash-swaption payoff
(4)	V_t	price under T-forward measure
(5)	V_t^{swp}	vanilla cash-swaption price (calibration target)
(6)	V_t^{cms}	CMS cap/floor price (alt. calibration target)
(7)	V_t^{idx}	vanilla index option price (calibration target)
(8)–(9)	F_{tT}, U_{tT}	local-vol dynamics under T-forward

Invariants and identities to verify:

1. *Cash annuity at par*: $\widehat{A}_T$ evaluated at $S_T = R_0^{swp}$ (i.e. ATM forward) should match the level consistent with the market discount curve to within a first-order correction; the gap is the cash-vs-market annuity difference standard in cash-settled-swaption pricing.
2. *Recovery of vanilla prices*: pricing (5) via the calibrated σ_F reproduces the input swaption smile by construction (Dupire); same for (7).
3. *Forward consistency*: F_{tT} is a $\mathbb{Q}^T$-martingale by construction, so $\mathbb{E}_t^T[F_{TT}] = F_{tT}$. Same for U_{tT}.

4. *Limit at $\rho = 0$*: with independent F and U, the hybrid price reduces to an iterated expectation over their marginals, computable in 1D once σ_F, σ_U are pinned.
5. *Strike intermediate values*: replacing U_T in the payoff by a constant K reduces (4) to (5), the vanilla cash-swaption, and the hybrid model and vanilla model must agree.
6. *Convexity adjustment consistency*: at $t = 0$, the difference between F_{0T} (the convexity-adjusted forward) and the spot forward swap rate S_0 should equal the Hagan / Pelsser convexity adjustment under matching assumptions.

5 Glossary of symbols (paper's notation)

U_t	Spot equity index level at t
U_{tT}	Forward price of U at t for delivery T
S_t	Forward swap rate at t with payment schedule $T_1, \ldots, T_n$
A_t	Risk-free annuity
$\widehat{A}_T$	Cash annuity (flat-curve discount on S_T)
θ	$+1$ payer / -1 receiver flag
$T = T_0$	Swaption exercise date
$T_1, \ldots, T_n$	Underlying swap payment dates
D_{tT}	Risk-free discount factor
F_{tT}	Convexity-adjusted forward swap rate, $\mathbb{E}_t^T[S_T]$
$\mathbb{Q}^T$	T-forward measure (numeraire D_{tT})
σ_F, σ_U	Local-volatility functions for F and U
$(W^F)^T, (W^U)^T$	Driving Brownian motions under $\mathbb{Q}^T$
ρ	Instantaneous correlation of $(W^F)^T$ and $(W^U)^T$
K	Strike for vanilla calibration instruments

TODO / Open questions

- Build a worked numerical example since the paper provides none: 1Y×10Y cash-swaption with strike $= U_T$ where U is a major equity index; benchmark against an iterated BS-style approximation as a sanity check.
- Calibration pipeline: implement steps (i)–(iv) on a synthetic market (flat smiles, synthetic CMS rates) to verify that vanilla prices are recovered exactly.
- Correlation sensitivity: produce a price-vs-ρ curve at plausible ATM vols ($\sigma_F, \sigma_U \sim 20\% - 30\%$) to gauge model risk on the unobservable input.
- Compare with a flat-vol Black approximation $V \approx D_{0T}\widehat{A}_0 \cdot \mathrm{BS}(F_0, U_0, \tilde{\sigma}, T)$ with $\tilde{\sigma}^2 = \sigma_F^2 - 2\rho\sigma_F\sigma_U + \sigma_U^2$.